Observer-Patch Holography and the Dark Sector

Modular Charge, the Anomalous Collar Source, and the Deep Galaxy Law

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Abstract

In observer-patch holography the Einstein equation sources curvature from the modular-charge stress of the screen under a universal coupling. Every screen source that carries modular charge therefore gravitates, whether or not it carries the electromagnetic readout, and the luminous-only Einstein relation holds exactly when the non-luminous charge vanishes. A dark sector is generic. The carrier is the anomalous modular energy on overlap collars, the same finite-stage remainder that the Einstein derivation controls: its rest-frame stress is bounded by the collar recovery defect, vanishes at full recovery, and dilutes as the inverse cube of the scale factor when conserved. The collars that carry it are galactic-scale objects, so a compact source sees only the ambient anomalous density; an isotropic ambient density has no quadrupole, and its tidal scale in the Solar System is four orders of magnitude below the Cassini bound. In the deep regime the anomalous density with linear enclosed mass gives the radial-acceleration relation $a = \sqrt{a_b a_0}$, flat rotation curves, and the baryonic Tully–Fisher relation $v^4 = GM_b a_0$; on the SPARC sample both relations are reproduced with one constant near $1.4 \times 10^{-10}\,\mathrm{m\,s^{-2}}$. The compact-source half of the collar-scale premise is a conditional theorem under an exponential recovery envelope, with a rate in the local record density, and the deep profile with its single constant is characterized as the unique law satisfying deep-regime scale covariance and quadrature composition of independent sources. All structural statements are machine-checked. The magnitude of the anomalous charge and the physical attachment of the envelope are not derived here.

Claim Boundary

The finite OPH results used here are the rest-frame Einstein relation on its named small-ball premises, the decomposition of the cap modular generator into a bulk part, a central part, and a collar-carried anomalous part, and the controlled bound on that anomalous part [1, 2]. Every theorem and proposition below is an algebraic or analytic consequence of those objects together with the premises named in the text, and each is checked in the Lean formalization [3]. Universal coupling and the collar-scale premise are structure fields or named hypotheses in that formalization; neither is established there. The linear enclosed-mass form is derived inside the formalization from two named premises, deep-regime scale covariance and quadrature source composition, which are themselves structure fields; the exponential recovery envelope behind the saturation theorem is the conditional mixing branch of the collar-recovery claim, taken as a hypothesis.

The paper does not supply a quotient-invariant receipt localizing the anomalous modular energy on collars, the value of the collar constant, the profile of the anomalous density outside the deep regime, a proof that the collars carrying the anomaly are galactic-scale, or a cosmological likelihood. The acceleration constant a_0 is a fitted comparison value. No cosmological result carries an OPH falsification verdict.

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1 What Sources Gravity in OPH

The Einstein arrow of the spacetime paper ends in

$$G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab} \rangle, \tag{1}$$

where $\langle T_{ab} \rangle$ is the local conserved stress reconstructed from positive null translations and modular charges of the record algebra [1]. In the rest frame the relation follows from three premises on a small geodesic ball of radius ℓ: generalized entropy is stationary, the bulk entropy variation equals $(8\pi^2\ell^4/15)\,\delta\langle T_{00}\rangle$, and the fixed-volume area variation equals $-(4\pi\ell^4/15)\,\delta G_{00}$. Exact arithmetic gives $\delta G_{00} = 8\pi G\,\delta\langle T_{00}\rangle$.

The stress entering this relation is modular charge. The screen does not consult the electromagnetic readout of a source before responding to it. That is the universal-coupling premise, and it is the only input the dark-sector question takes from the gravity derivation.

2 The Generic Dark Sector

Let a finite family of screen sources be indexed by i, each with rest-frame modular-charge contribution q_i and a flag recording whether it carries the electromagnetic readout. Write $Q_{\rm lum}$ for the sum over flagged sources and $Q_{\rm dark}$ for the sum over the others, so that $Q_{\rm tot} = Q_{\rm lum} + Q_{\rm dark}$.

Premise 2.1 (Universal coupling)**.** *The stress contraction entering the rest-frame Einstein relation equals $Q_{\rm tot}$, with no readout-dependent weight.*

Theorem 2.2 (Geometry responds to all modular charge)**.** *Under Premise 2.1, $\delta G_{00} = 8\pi G\,(Q_{\rm lum}+Q_{\rm dark})$.*

Theorem 2.3 (The luminous-only relation)**.** *Under Premise 2.1, $\delta G_{00} = 8\pi G\, Q_{\rm lum}$ holds if and only if $Q_{\rm dark} = 0$.*

Corollary 2.4 (A geometric excess forces a dark source)**.** *If $\delta G_{00} \neq 8\pi G\, Q_{\rm lum}$, some source without the electromagnetic readout carries nonzero modular charge.*

Proposition 2.5 (Attractive sign)**.** *If every non-luminous charge is nonnegative, $Q_{\rm dark} \geq 0$, and the dark contribution enters the weak-field equation with the sign of baryonic mass.*

Proofs. $Q_{\rm tot} = Q_{\rm lum} + Q_{\rm dark}$ is a finite-sum identity. Theorem 2.2 substitutes it into the rest-frame relation. Theorem 2.3 cancels $8\pi G > 0$. The corollary is the contrapositive of the sum of zeros, and the proposition is a sum of nonnegative terms. □

In a theory where gravity is sourced by modular charge, the existence of a dark sector is settled by these four lines: any geometric excess over the luminous prediction is by itself the presence of non-luminous modular charge. The remaining questions are which screen object carries that charge, how much of it there is, and how it is distributed.

3 The Modular-Anomaly Source

On the support-visible branch the cap modular generator has the local form

$$K_C = 2\pi B_C + K_C^{\rm anom} + \text{const}, \tag{2}$$

with B_C the bulk boost generator and $K_C^{\rm anom}$ the nonadditive part carried on the overlap collar [1]. At finite regulator stage the Einstein derivation carries this part as a remainder and bounds it:

$$|\delta\langle E_C^{(\delta)}\rangle| \leq C_C\,\eta_\delta, \tag{3}$$

where η_δ is the collar recovery defect and C_C depends only on the collar model and the bounded observable class.

Definition 3.1 (Anomalous stress and density)**.**

$$\delta\langle T_{00}^{\rm anom}\rangle := \frac{15}{8\pi^2\ell^4}\,\delta\langle E_C^{(\delta)}\rangle, \qquad \rho_A := \frac{1}{c^2}\,\delta\langle T_{00}^{\rm anom}\rangle. \tag{4}$$

The dark-sector identification is that this anomalous modular energy is the non-luminous charge of Section 2. It carries no electromagnetic readout by construction: it is the part of the modular generator that no local record exposes. Three consequences follow from (3) and (4).

Theorem 3.2 (Bounded source)**.** $|\delta\langle T_{00}^{\mathrm{anom}}\rangle| \le 15\, C_C\, \eta_\delta/(8\pi^2\ell^4)$.

Theorem 3.3 (Full recovery switches the source off)**.** *If* $\eta_\delta = 0$ *then* $\delta\langle T_{00}^{\mathrm{anom}}\rangle = 0$. *Along any family with fixed* ℓ*, bounded* C_C*, and* $\eta_\delta \to 0$*, the anomalous stress tends to zero.*

Theorem 3.4 (Cold scaling)**.** *A conserved anomalous charge in a comoving cell of physical volume* $\propto a^3$ *has density* $\propto a^{-3}$*.*

Proofs. Theorem 3.2 multiplies (3) by the positive kernel inverse. Theorem 3.3 is the case $\eta_\delta = 0$ of the bound and a squeeze along the family. Theorem 3.4 is $\rho\, a^3 = Q$. □

In the weak-field limit the anomalous density enters the Poisson equation beside the baryons,

$$\nabla^2\Phi = 4\pi G\,(\rho_b + \rho_A), \tag{5}$$

with the attractive sign of Proposition 2.5 whenever the anomalous charge is nonnegative.

4 Galactic-Scale Collars and the Solar System

The collars in (2) are overlap collars between observer patches, and the recovery defect η_δ measures how far the repaired record on a collar falls short of determining its complement. Where records are dense and repair saturates, the defect is small; where the screen is sparsely recorded, over the settled outskirts of a galaxy, the defect is carried on long-lived cuts. The anomalous charge is therefore a property of the record state of a region, and the region in question is the galaxy.

Premise 4.1 (Collar scale)**.** *The collars carrying* $\delta\langle E_C^{(\delta)}\rangle$ *are galactic-scale settled cuts. A compact source inside the galaxy does not generate anomalous charge of its own; it sits in the ambient anomalous density* ρ_A *of its neighbourhood.*

The compact-source half of this premise is a conditional theorem. The collar-recovery claim of the spacetime paper [1] bounds the defect on its conditional mixing branch by an exponential envelope, which we write in the local record density ρ as

$$\eta_\delta \le \kappa\, B\, \exp\Bigl(-\frac{\delta\rho^{1/3} - r_0}{\zeta}\Bigr), \tag{6}$$

with κ the mixing prefactor, B the boundary size in UV cells, r_0 an offset, ζ the decay constant, and $\delta\rho^{1/3}$ the collar depth counted in UV record spacings.

Theorem 4.2 (Compact-source saturation)**.** *Under the envelope* (6)*, along any family of collars with fixed radius* ℓ *and bounded collar constant* C_C*, the recovery defect and the anomalous stress tend to zero as the depth grows at fixed record density, and as the record density grows at fixed positive depth. Quantitatively,*

$$|\delta\langle T_{00}^{\mathrm{anom}}\rangle| \le \frac{15\, C_C}{8\pi^2\ell^4}\,\kappa B\, \exp\Bigl(-\frac{\delta\rho^{1/3} - r_0}{\zeta}\Bigr), \tag{7}$$

and the envelope bound falls below ε *exactly when* $\delta\rho^{1/3} > r_0 + \zeta\log(\kappa B/\varepsilon)$*. Where records are dense, a source generates no persistent anomalous charge of its own.*

The delimitation is exact in both directions and machine-checked. The identically zero defect satisfies every envelope hypothesis, so the envelope never forces a nonzero defect: that the galactic-scale settled cuts carry an order-one defect is a premise, not a consequence. And an explicit sparsely recorded instance sits at the bound with an order-one defect and strictly positive stress, so the same hypotheses that screen dense regions admit a nonzero dark source on sparse cuts. What is not established is that physical collars satisfy (6): the envelope is the conditional mixing branch of the collar-recovery claim, taken here as a hypothesis.

Under Premise 4.1 the Solar System is a test of the ambient density alone.

Proposition 4.3 (An isotropic ambient density has no quadrupole)**.** *The potential of a locally uniform density is $(2\pi G\rho_A/3)\,r^2$. Its projection on the quadrupole weight $P_2(\cos\theta)$ vanishes, $\int_{-1}^{1}(\mu^2 - \frac{1}{3})\,\mathrm{d}\mu = 0$, so the quadrupole coefficient Q_2 is zero. The only footprint is the isotropic tidal scale $4\pi G\rho_A/3$.*

With the local dark density $\rho_A \simeq 6.8\times10^{-22}\,\mathrm{kg\,m^{-3}}$, the value inferred from Galactic kinematics [10], the tidal scale is

$$\frac{4\pi G\rho_A}{3} = 1.9\times10^{-31}\,\mathrm{s}^{-2}, \tag{8}$$

against the Cassini quadrupole bound $Q_2 = (1.6\pm1.8)\times10^{-27}\,\mathrm{s}^{-2}$ [9]: four orders of magnitude below the uncertainty, with zero quadrupole. The anomalous mass enclosed within ten astronomical units is $5\times10^{-15}\,M_\odot$. Both numbers are machine-checked inequalities in the formalization. The Solar System constrains the premise of Section 4 and nothing else in the construction.

5 The Deep-Regime Profile

Around a baryonic mass M_b, the codimension-one collar count of the microphysics paper [2] gives the number of settled cuts enclosing the source within radius r as proportional to r. Rather than positing the enclosed-mass profile as a functional form, the deep regime is characterized by two premises, each independently falsifiable.

Premise 5.1 (Deep-regime scale covariance)**.** *In the deep regime the enclosed anomalous mass is degree-one homogeneous in the radius: $M_A(M_b,\lambda r) = \lambda\,M_A(M_b,r)$ for every $\lambda>0$. A sparsely recorded deep cut retains no length scale, so rescaling the radius rescales the enclosed mass by the same factor.*

Premise 5.2 (Quadrature composition)**.** *The anomalous amplitudes of independent baryonic sources compose in quadrature: $M_A(m_1+m_2,r)^2 = M_A(m_1,r)^2 + M_A(m_2,r)^2$. The collar defects of distinct sources are independent, their variances add, and the gravitating anomalous charge is the amplitude of the unrecovered record fluctuation.*

Theorem 5.3 (Profile characterization)**.** *Let $M_A(M_b,r)$ be nonnegative, monotone in the source, somewhere nonzero, and satisfy Premises 5.1 and 5.2. Then there is a unique constant $a_0>0$ with*

$$M_A(r) = r\sqrt{\frac{M_b a_0}{G}}, \qquad \rho_A(r) = \frac{1}{4\pi r^2}\sqrt{\frac{M_b a_0}{G}}. \tag{9}$$

Proof. Scale covariance factorizes $M_A(M_b,r) = r\,f(M_b)$. Quadrature composition makes f^2 additive on positive sources, monotonicity makes it monotone, and a function additive and monotone on the positive reals is linear: $f^2 = \lambda M_b$ with $\lambda > 0$ by nondegeneracy. Setting $a_0 = \lambda G$ gives (9); uniqueness follows by evaluating at $M_b = r = 1$. Each premise is load-bearing: an additive law $M_A \propto rM_b$ satisfies everything except quadrature composition, and an r^2 law everything except scale covariance. $\square$

Corollary 5.4 (Linear enclosed mass)**.** *The deep-regime density is the shell derivative of the enclosed mass, $M'_A(r) = 4\pi r^2 \rho_A(r)$, with M_A linear in r and the coefficient $\sqrt{M_b a_0/G}$ forced by Theorem 5.3.*

Deep-limit scale invariance is the known symmetry of the modified-dynamics limit [7], and square-root mass laws appear in entropic-gravity proposals [8]. The characterization here derives the full profile with its unique constant from the two premises jointly, and it reads the square root as the variance additivity of independent collar defects under source composition.

Premise 5.2 itself follows from a more primitive per-cut model. Suppose the nr nested cuts within radius r each carry anomalous amplitude $\sqrt{V(M_b)}$, the standard deviation of the collar defect sourced by the baryons, with the variances of independent sources adding and the variance monotone in the source. Then the variance function is forced linear, $V(M_b) = cM_b$ with a unique $c > 0$, quadrature composition is a theorem rather than a premise, the induced law satisfies Theorem 5.3, and its constant is exactly

$$a_0 = G\, n^2 c. \tag{10}$$

The dictionary is exact but determines only the product n^2c: two models with different cut densities and equal n^2c induce the same enclosed-mass law, machine-checked. The magnitude question is therefore the value of the collar density and of the per-cut variance constant; neither is derived here.

Theorem 5.5 (Deep radial-acceleration relation)**.** *The anomalous acceleration of the profile (9) is*

$$a_A(r) = \frac{GM_A(r)}{r^2} = \frac{\sqrt{GM_b a_0}}{r} = \sqrt{a_b(r)\, a_0}, \qquad a_b = \frac{GM_b}{r^2}. \tag{11}$$

Theorem 5.6 (Flat rotation curve and baryonic Tully–Fisher relation)**.** *The circular speed supported by a_A satisfies $v^2 = a_A r = \sqrt{GM_b a_0}$, independent of r, and hence*

$$v^4 = GM_b\, a_0. \tag{12}$$

Proofs. $GM_A/r^2 = G\sqrt{M_b a_0/G}/r = \sqrt{GM_b a_0}/r$, and $\sqrt{GM_b a_0}/r = \sqrt{(GM_b/r^2)a_0}$ for $r > 0$. Multiplying by r removes the radius; squaring gives (12). $\square$

The exponents 1/2 in (11) and 4 in (12) are the parameter-free content of the profile. The constant a_0 collects C_C, the collar density, and the small-ball normalization; none of these is fixed by the source, so a_0 enters every comparison as a fitted value. The deep regime is the range $a_b \ll a_0$, equivalently $r \gg r_M = \sqrt{GM_b/a_0}$. At the matching radius both accelerations equal a_0 exactly, and the anomalous acceleration dominates the baryonic one precisely outside r_M; both statements are machine-checked. Inside r_M the enclosed anomalous mass is small compared with M_b and the response is Newtonian; the profile through the transition is not fixed by Premises 5.1 and 5.2.

6 Lensing and the Relativistic Source

The anomalous charge enters (1) through the same $\langle T_{ab} \rangle$ that sources the timelike response. Under Premise 2.1 there is one stress tensor, and the null directions of the spacetime paper's tomography frame read the same charge as the timelike ones [1].

Proposition 6.1 (One source for lensing and dynamics)**.** *Under Premise 2.1, the mass inferred from null geodesics through a region equals the mass inferred from timelike orbits in it: both are* $Q_{\mathrm{lum}} + Q_{\mathrm{dark}}$*.*

Proof. Both readouts are contractions of the one tensor in (1) with the respective tangent vectors, and the premise supplies no readout-dependent weight. $\square$

The proposition is structural. No relativistic refinement limit is supplied that turns the weak-field density (9) into a full T_{ab}^{anom} with its pressure and anisotropic stress; consequently, the lensing statement is that the anomalous charge lenses with the same weight with which it moves stars.

7 Comparison with Rotation-Curve Data

Theorems 5.5 and 5.6 are compared with the SPARC sample of 175 disk galaxies with $3.6\,\mu$m photometry [5] under the standard cuts of McGaugh, Lelli, and Schombert [6]: quality flag at most 2, inclination at least 30°, velocity error at most ten percent, stellar mass-to-light ratios 0.5 for disks and 0.7 for bulges, leaving 2700 points in 149 galaxies. The computation is released with the paper [4].

Radial acceleration. On the self-consistent deep subset $g_{\mathrm{bar}} < f a_0$, the fixed-exponent fit $g_{\mathrm{obs}} = \sqrt{g_{\mathrm{bar}} a_0}$ gives

$$a_0 = 1.57 \times 10^{-10}, \quad 1.40 \times 10^{-10}, \quad 1.28 \times 10^{-10}\ \mathrm{m\,s^{-2}}$$

for $f = 0.3$, 0.1, and 0.03, with 1651, 1043, and 233 points and rms scatter between 0.14 and 0.17 dex. The free exponents on the same subsets are 0.60 ± 0.01, 0.57 ± 0.02, and 0.39 ± 0.08, with point-bootstrap errors. The exponent moves toward 1/2 as the cut deepens and the sample runs out of points before the deep regime is clean; the data are consistent with exponent 1/2 at the available depth, with the constant between 1.3 and $1.6 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ depending on the cut.

Baryonic Tully–Fisher. For the 123 galaxies with a measured flat velocity that pass the same cuts, with $M_b = 0.5 L_{[3.6]} + 1.33 M_{\mathrm{HI}}$, the fixed-exponent normalization $a_0 = v_f^4/(G M_b)$ gives $a_0 = 1.55 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ with 0.26 dex scatter, and the free exponent from an ordinary least-squares fit of $\log v_f$ on $\log M_b$ is 3.75 ± 0.10.

One constant. Theorems 5.5 and 5.6 share one a_0. The constants recovered from the two relations differ by 0.046 dex at $f = 0.1$, inside the combined scatter. This is the one nontrivial check the deep profile admits with a single fitted constant, and it passes.

8 Cosmological Scaling

If the total anomalous charge in a comoving region is conserved, which holds whenever the collar family of the region is stable under the expansion, Theorem 3.4 gives $\rho_A \propto a^{-3}$: the anomalous medium is pressureless on cosmological scales. With Premise 4.1 it clusters with the galaxies whose collars carry it. The abundance Ω_A, the perturbation initial data, and the transfer functions require the collar density of the early screen and are not derived here; the scaling is the statement that the medium is cold.

9 Required Physical Constructions

1. **Collar scale.** The compact-source half of Premise 4.1 is Theorem 4.2, conditional on the exponential envelope (6). What remains is the physical attachment: a proof that physical collars satisfy the mixing branch behind the envelope, and a positive mechanism for the order-one defect on galactic-scale settled cuts, which the envelope provably does not force.
2. **Magnitude.** The normalization of $\delta\langle E_C^{(\delta)}\rangle$ to repair conservation on the Ward/Bianchi branch is not derived. The per-cut dictionary (10) reduces the question to the collar density n and the per-cut variance constant c: a repair-conservation normalization would fix c, and the codimension-one collar count of the microphysics paper would fix n.
3. **Profile.** Theorem 5.3 reduces the profile to Premises 5.1 and 5.2; what remains is deriving those two premises from the collar geometry of a point source, and extending the profile through the transition to the Newtonian regime.
4. **Localization.** No quotient-invariant receipt is supplied that identifies the collar operator in (2) with the small-ball remainder in (4).
5. **Abundance.** The comoving anomalous charge of the early screen, the resulting Ω_A, and the joint galaxy, cluster, lensing, CMB, and growth likelihood are not derived.

10 Conclusion

Under universal coupling the screen gravitates according to its total modular charge, so a dark sector is a theorem-level feature of OPH: any geometric excess over the luminous prediction is the presence of non-luminous modular charge. The anomalous modular energy on overlap collars is the carrier the Einstein derivation itself supplies. It is bounded by the recovery defect, absent at full recovery, cold when conserved, and, on galactic-scale collars, invisible to the Solar System by four orders of magnitude. Its deep-regime profile gives the radial-acceleration relation, flat rotation curves, and the baryonic Tully–Fisher relation with one constant, and the SPARC sample returns that constant consistently from both relations. The compact-source half of the collar-scale premise is a conditional saturation theorem with an exponential rate in the local record density, and the profile with its unique constant is characterized by deep-regime scale covariance and quadrature composition of independent sources. The physical attachment of the envelope, the mechanism selecting galactic-scale cuts, and the magnitude of the charge are the work that turns the remaining premises of Sections 4 and 5 into derivations.

References

[1] B. Mueller et al., *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency*, OPH spacetime and Einstein paper. https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/recovering_observer_spacetime_and_einstein_dynamics_from_overlap_consistency.pdf

[2] B. Mueller et al., *Federated Echosahedral Screen Microphysics: Patch Hardware, Records, and Observer Synchronization in OPH*. https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/screen_microphysics_and_observer_synchronization.pdf

[3] B. Mueller et al., *Observer-Patch Holography: machine-checked Lean formalization*, Einstein-branch modules `SmallBall`, `DarkSector`, `CollarScaleSaturation`, `DeepProfileClosure`, and `PerCutCollarComposition`, 2026. https://github.com/FloatingPragma/observer-patch-holography/tree/main/Lean/ObserverPatchHolography/EinsteinBranch

[4] B. Mueller, *Deep-regime comparison of the OPH galaxy law with SPARC*, 2026. https://github.com/FloatingPragma/observer-patch-holography/tree/main/code/cosmology/rar_deep_regime

[5] F. Lelli, S. S. McGaugh, and J. M. Schombert, *SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves*, Astron. J. 152, 157 (2016), arXiv:1606.09251. https://arxiv.org/abs/1606.09251

[6] S. S. McGaugh, F. Lelli, and J. M. Schombert, *Radial Acceleration Relation in Rotationally Supported Galaxies*, Phys. Rev. Lett. 117, 201101 (2016), arXiv:1609.05917. https://arxiv.org/abs/1609.05917

[7] M. Milgrom, *The MOND limit from spacetime scale invariance*, Astrophys. J. 698, 1630 (2009), arXiv:0810.4065. https://arxiv.org/abs/0810.4065

[8] E. Verlinde, *Emergent Gravity and the Dark Universe*, SciPost Phys. 2, 016 (2017), arXiv:1611.02269. https://arxiv.org/abs/1611.02269

[9] R. S. Park et al., *Improved constraints on modified Newtonian gravity from Cassini radio tracking data*, 2026, arXiv:2602.17884. https://arxiv.org/abs/2602.17884

[10] P. F. de Salas and A. Widmark, *Dark matter local density determination*, Rep. Prog. Phys. 84, 104901 (2021), arXiv:2012.11477. https://arxiv.org/abs/2012.11477

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